SussexBudgetProductions

Module: Data Science Research Methods

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INTRODUCTION

SussexBudgetProductions recently faced significant losses with a comedy-action-thriller, prompting a shift in strategy. The team have decided to focus on either Romance or Horror, under the assumption that higher IMDb scores align with profitability. This report evaluates IMDb data to identify which genre holds better potential for profit and recommends the most suitable director and lead actor.

The analysis incorporates IMDb score distributions, review counts, and profit-to-budget ratios. Data cleaning, feature engineering, exploratory analysis, and hypothesis testing were conducted to deliver data-driven recommendations for optimal genre, director, and cast choices.

METHODS

1. Data Collection and Preparation

The dataset contains 5,043 movies with 26 features

- **Key insights:**
  - **Gross Revenue:** Average of £48.5M, peaking at £760M.
  - **IMDb Rating:** Average rating of 6.44, ranging from 1.6 to 9.5.
  - **Genres:** "Drama" is the top genre with 236 movies.

- **Dataset Overview:**
  - **Missing Values:** Some columns have missing data.
  - **Irrelevant Columns:** Certain columns are unnecessary for the analysis.
  - **Unreadable Column Names:** Some columns need clearer names.

2. Data Wrangling and Cleaning

- **Dropping Unnecessary Columns:** Columns such as '*colour', 'movie_imdb_link', 'duration'*, kind of columns were excluded due to their minimal impact on the analysis and secondary influence on movie success. Additionally, *actor_2_name', 'actor_3_name'* were removed as '*actor_1_name'* is the primary focus.
- **Renaming the columns**: Standardized column names to enhance readability.

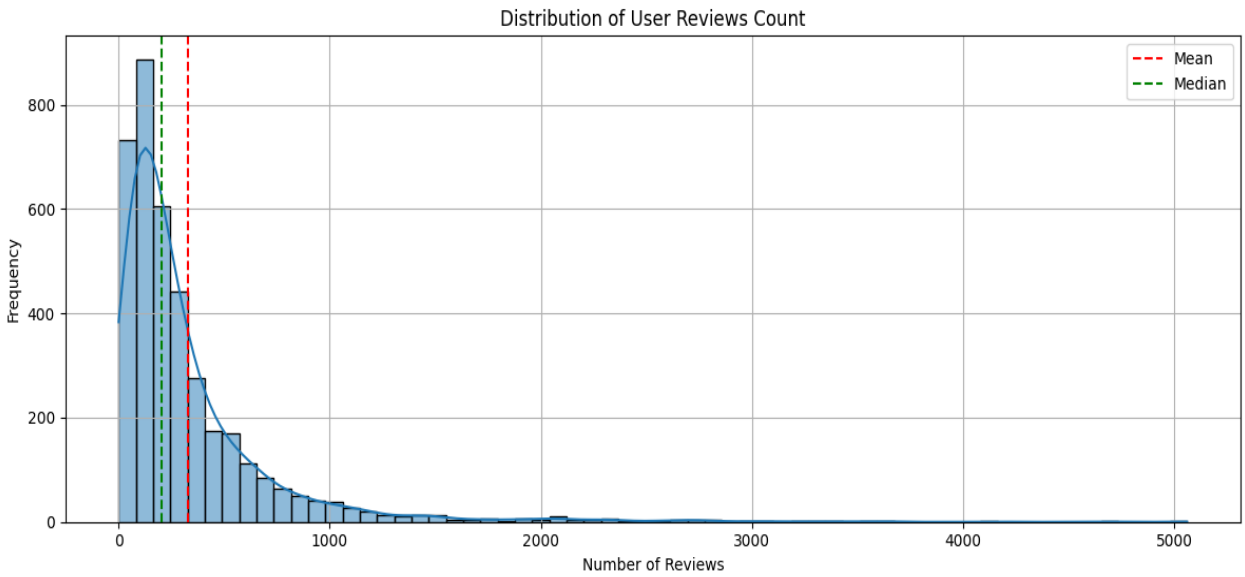
- **Handling Missing Values & Duplicate Rows**: Rows with missing values in critical columns were removed, resulting in the exclusion of **1,155** rows to ensure data accuracy. Additionally, **35** duplicate rows were identified and removed, to maintain dataset integrity.
- **No outlier treatment**: Outliers were retained as they represent genuine cases (e.g., blockbuster or low-budget films) rather than errors. Removing them could misrepresent the true variability and profitability range across genres.
- **New Features Creation**:
  - ***'new_genre'*** - categorizing movies as "Romance," "Horror," or "Other".
  - ***'Profit_to_Budget_Ratio'* -** Developed to evaluate profitability relative to budget. This ratio, **unaffected by currency value changes over time**, offers a consistent profitability measure across the dataset's 1916–2016 range.

3. Data Filtering and Exploratory Data Analysis (EDA):

- **Filtering based on User Reviews count:**
  - To ensure reliable IMDb scores, filter the data to include movies with a high number of reviews (e.g., above the median). This reduces biases from films with fewer reviews.

Summary statistics:

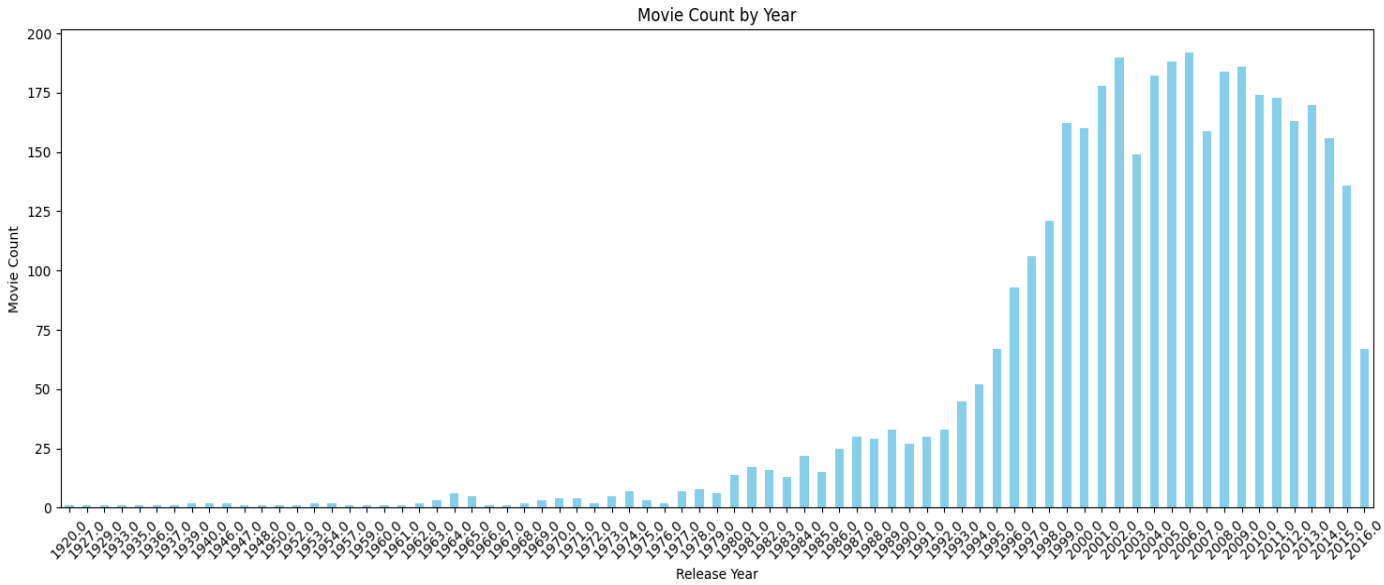
```
count    3853.000000
mean      326.687776
std       407.941808
min         1.000000
25%       102.000000
50%       203.000000
75%       391.000000
max      5060.000000
```



- The user reviews distribution is **heavily right-skewed**, showing most movies have few reviews.
- To ensure reliable analysis, only movies with **400+ reviews (25th percentile)** were included. This helps mitigate biases from movies with fewer reviews.

➢ **Filtering based on Release Year:**

- Given dataset spans from 1916 to 2016, focusing on recent films (**1995 onward**) aligns better with current trends and audience preferences, providing more accurate insights into contemporary tastes. This reduces the influence of older films.



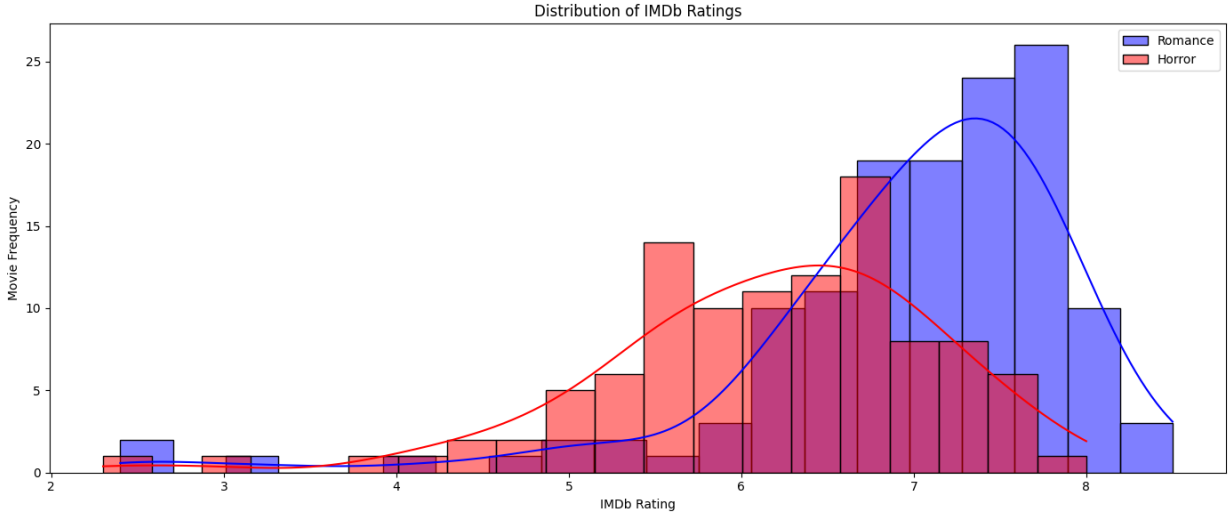
- The data shows a notable rise in movie production since the 1990s, with a sharp increase in releases.

➢ **Filtering Romance and Horror Movies:**

- Only movies labelled as either *"Romance" or "Horror"* were retained for further analysis, aligning with company interests.

➢ **Genre Distribution**:

- To understand the dataset's genre composition, the frequency and average IMDB score of each genre were calculated and graphed.



- Romance movies generally have higher, more consistent ratings, while Horror films show greater variability. Based on the IMDB score graph, the **Romance genre seems preferable**, but this will be confirmed through hypothesis testing.

➢ **Further Analysis:**

- The **Drama** genre is the most promising, supported by both graphical and categorical data. ***The Dark Knight*** stands out with an IMDb rating of ***9***.
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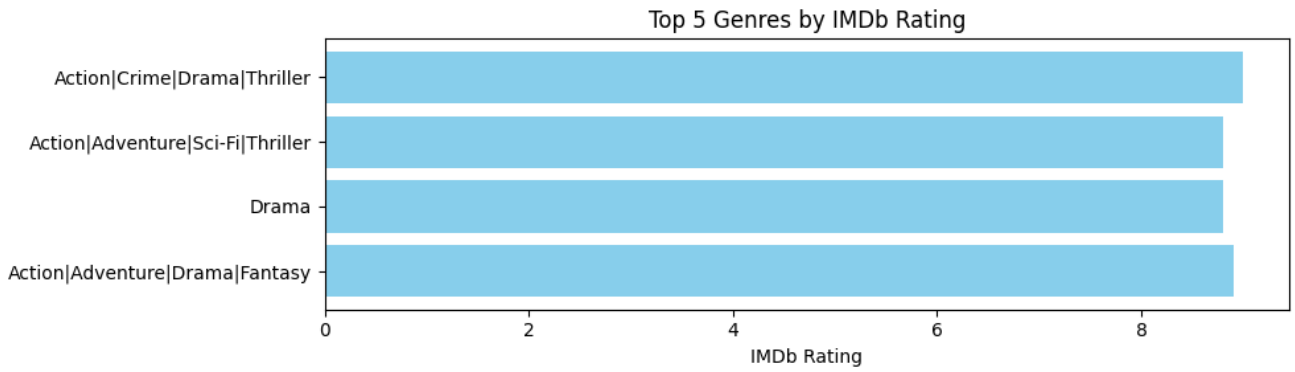

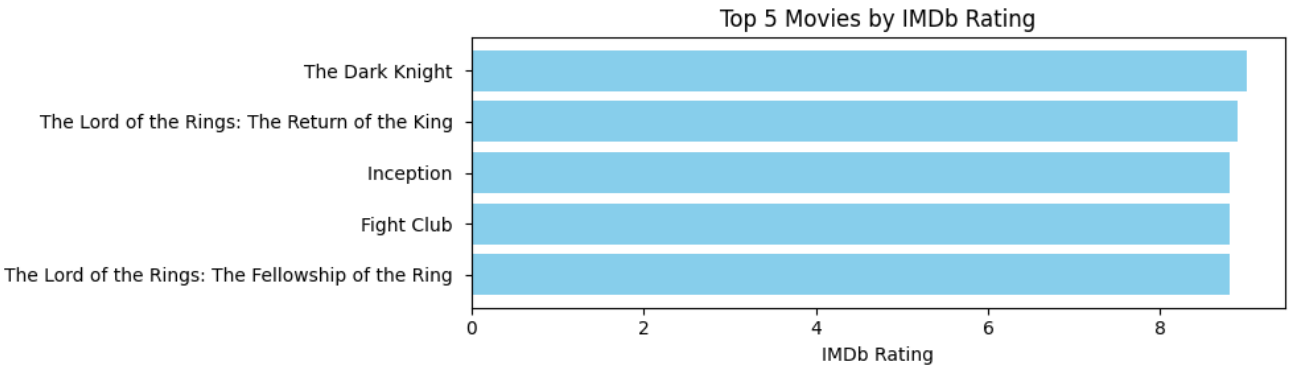

4. Hypothesis Testing

To identify the most profitable genre between Romance and horror, IMDb ratings and the profit-to-budget ratio are analysed. A higher IMDb rating suggests greater audience appeal, while the profit-to-budget ratio indicates a film's financial success.

- **T-test for Profit-to-Budget Mean:**
  - The p-value of **0.2050** indicates **no significant difference** in profit-to-budget ratios between the genres likely due to similar production cost structures balancing profitability, leading to the conclusion that the null hypothesis cannot be rejected at the 5% level.
- **T-test for IMDb Ratings Mean:**
  - The p-value of **0.000** indicates **significant difference** in IMDb ratings suggests one genre has higher ratings, helping decide the genre based on IMDb

RESULTS

- **Best Genre Analysis:**

  Based on the comparison of average IMDb ratings, the Romance genre has a higher average rating (**6.97**) compared to the Horror genre (**6.15**). Therefore, **Romance is a better choice for the next movie.**

- **Best Director Analysis**:

  After *filtering directors with more than two romance films*, average IMDb ratings were calculated for each. Averaging ratings from a single film may not ensure reliable filmmaking quality. The top directors based on IMDb ratings are:

director_name	imdb_rating
Ridley Scott	8.5
Jean-Pierre Jeunet	8.4
Michel Gondry	8.3
Tim Miller	8.1
Richard Linklater	8.0
Michel Hazanavicius	8.0
Danny Boyle	8.0
Stephen Chbosky	8.0
Spike Jonze	8.0
James Mangold	7.9

These directors consistently deliver high-rated films in Romance, with **Ridley Scott** as the top choice at an IMDb rating of **8.5**.

- **Best Actor Analysis:**

  The analysis assesses lead actors in romance films by **IMDb ratings and Facebook likes**, showing audience reception and social media appeal, which can boost movie visibility. Top actors with high IMDb ratings and strong social media followings are strong candidates for the next film.

actor_name	imdb_rating	actor_facebook_likes
Djimon Hounsou	8.5	3000.0
Mathieu Kassovitz	8.4	326.0
Ryan Reynolds	8.1	16000.0
Logan Lerman	8.0	8000.0
Bérénice Bejo	8.0	996.0
Anil Kapoor	8.0	668.0
Vernon Dobtcheff	8.0	50.0
Ryan Gosling	7.9	33000.0
Sandra Ellis Lafferty	7.9	523.0
Chen Chang	7.9	103.0

Djimon Hounsou stands out as the top actor with an IMDb rating of **8.5**. However, romance films often need actors with broader appeal, making **Ryan Reynolds** a strong alternative. His IMDb rating of **8.1** remains solid, and with **16,000** Facebook likes, his extensive social media following can enhance visibility and draw a larger audience. This makes him a compelling choice for a lead role in a romance film.

CONCLUSION

This report suggests producing a **Romance** movie directed by **Ridley Scott** and starring **Ryan Reynolds**. The decision is based on the higher average **IMDb rating** for Romance films, indicating favorable audience reception. Both Scott and Reynolds have proven track records of successful, high-rated productions. By leveraging this data, SussexBudgetProductions can aim for profitability and strengthen its position in the film industry.